

Math 472: Homework 09

Due Monday, April 20

Problem 1 (Exercise 10.6). We are interested in testing whether a coin is balanced based on the number of heads Y in 36 flips; that is, $H_0 : p = .5$ vs $H_a : p \neq .5$. If we use the rejection region $|y - 18| \geq 4$, what is

- (a) the value of α ?
- (b) the value of β if $p = .7$?

Problem 2. Please answer “true” or “false” to the following questions.

- (a) The level of the test computed in Problem 1(a) is the probability that H_0 is true.
- (b) The value of β in Problem 1(b) is the probability that H_a is true.
- (c) In Problem 1(b), β was computed assuming the null hypothesis was false.
- (d) If β was computed when $p = .55$, the value would be larger than the value of β obtained in Problem 1(b).
- (e) The probability that the test mistakenly rejects H_0 is β .
- (f) Suppose the rejection region was changed to $|y - 18| \geq 2$.
 - (i) This RR would lead to rejecting the null hypothesis more often than the RR used in Problem 1.
 - (ii) If α was computed using this new RR, the value would be larger than the value obtained in Problem 1(b).
 - (iii) If β was computed when $p = .7$ and using this new RR, the value would be larger than the value obtained in Problem 1(b).

Problem 3 (Exercise 10.106). A survey of voter sentiment was conducted in four political wards to compare the fraction of voters favoring candidate Jones. Random samples of 200 voters were polled in each of the four wards, with the results as shown in the following table:

Opinion	Ward				Total
	1	2	3	4	
Favor A	76	53	59	48	236
Do not favor A	124	147	141	152	564
Total	200	200	200	200	800

The number of voters favoring Jones in the four samples can be regarded as four independent binomial random variables X_1, X_2, X_3 , and X_4 , of the form $X_i \sim \text{Binomial}(200, p_i)$ for $i = 1, 2, 3, 4$, where p_i is the proportion of voters in ward i who support Jones.

In the following parts we will construct a likelihood ratio test of the hypothesis that the fraction of voters favoring candidate Jones are the same in all four wards.

- (a) State the null and alternative hypotheses.
- (b) Describe the null hypothesis geometrically. What is the dimension of the null hypothesis?
- (c) Write down the joint likelihood of X_1, X_2, X_3 and X_4 and find its maximum.
- (d) Write down the joint likelihood of X_1, X_2, X_3 and X_4 when H_0 is true, and find its maximum.
- (e) Using your answer to the previous two parts, compute Λ as it was defined in lecture:

$$\Lambda = -2 \log \left(\frac{\sup_{\theta \in \Theta_0} L(\theta)}{\sup_{\theta \in \Theta} L(\theta)} \right)$$

- (f) Compute a p -value for the data, and interpret your result in words.

Problem 4 (Exercise 10.88). Refer to problem 4 in Homework 8. Find the power of the test for each alternative in (a) – (d):

- (a) $H_a : p = .4$
- (b) $H_a : p = .5$
- (c) $H_a : p = .6$
- (d) $H_a : p = .7$
- (e) Sketch a graph of the power function.

Problem 5 (Method of Moments - exercise 9.69). Let X be the proportion of allotted time that a randomly-selected student spends working on an exam. Suppose that X is a random variable with pdf

$$f(x) = \begin{cases} (\theta + 1)x^\theta & : 0 \leq x \leq 1 \\ 0 & : \text{otherwise} \end{cases}$$

where $\theta > -1$ is an unknown parameter. A random sample of 10 students yields data

$$\begin{aligned} x_1 = .92, x_2 = .79, x_3 = .90, x_4 = .65, x_5 = .86 \\ x_6 = .47, x_7 = .73, x_8 = .97, x_9 = .94, x_{10} = .77 \end{aligned}$$

- (a) Use the method of moments to obtain an estimator of θ and then compute the estimate for this data.
- (b) Use the factorization criterion to show that the statistic $T(X_1, \dots, X_n) = -\sum_{i=1}^n \log(Y_i)$ is sufficient for estimating θ .
- (c) Is the method of moments estimator a function of T ? What implications does this have?

Problem 6 (Exercise 9.19). Let $Y_1, \dots, Y_n$ be a random sample from the pdf

$$f(x) = \theta y^{\theta-1} \mathbf{1}_{[0 < y < 1]},$$

where $\theta > 0$. Show that $\bar{Y}$ is a consistent estimator of $\frac{\theta}{\theta+1}$.